ENGINEERING BRIEF · AGENT A · FEATURE

Roomio — Persona Rooms & Smart Suggestions

Two linked features: (1) ten Pinterest-grounded "start from who you are" room presets — default furnished rooms for personas like bachelor, anime fan, young couple, family-with-kids; and (2) a live design-suggestion engine that flags missing essentials (a bed, a rug, lighting layers) as confirmable dropdown prompts as the user builds.

PINTEREST-GROUNDED · PERSONA PRESETS · SUGGESTION RULEBOOK

Repo: `~/Desktop/personal/roomio` (front-end) · **Agent:** A
Coordination: `roomio.txt` · **Date:** June 29, 2026
Companion to: Room Designer v2, Detection & Camera briefs (Agent B)

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1. The Feature & a Hard Rule on Sourcing

> This brief turns Roomio from "empty box you furnish" into "start from a vibe that fits who you are, then get nudged toward a complete, well-designed room." Two features: **persona presets** (pre-furnished starting rooms) and a **suggestion engine** (live "you're missing X" dropdowns). Both must be grounded in real design patterns, not invented taste.

NON-NEGOTIABLE SOURCING RULE

Do **not** design these from intuition. Both features must be grounded in **real Pinterest pins and documented design sources**. For every genre, you (Agent A) must actually look at Pinterest boards and reference images for that aesthetic, identify the *recurring* elements across many pins (the items that appear again and again), and build the preset from those common patterns — not from a guess about what the genre "should" look like. The seed research in this brief gets you started; your visual pattern-matching makes it authentic.

Why this matters: a "bachelor pad" or "anime room" that comes from a guess will feel generic and wrong to the exact users who chose it. These users *know* the aesthetic intimately. The credibility of the whole feature rests on the presets looking like they came off the user's own Pinterest board.

METHOD IN ONE LINE

For each genre: open Pinterest + reference sources → collect 15–30 representative rooms → tally which furniture, colors, materials, and decor recur → encode the recurring set as the preset. Frequency across pins is the signal.

2. Feature 1: Persona Room Presets

A new entry point alongside the blank-room wizard: **"Start from a style that's you."** The user picks a persona/genre; Roomio loads a pre-furnished room (structure + placed archetypes + wall/floor materials + decor) that they then customize. It's a head start, not a lock-in — everything remains editable.

What a preset contains

- A room **shape + dimensions** (sensible default for the persona — e.g. a studio footprint for "bachelor," a larger family room for "family-with-kids").
- A **wall color + floor material** drawn from the genre's recurring palette.
- A set of **placed furniture archetypes** (from Roomio's existing corpus) in a sensible layout.
- **Decor / accent** items that signal the genre (the recurring "signature" pieces).
- A short **style note** describing the vibe, so the user understands what they picked.

CONSTRAINT: USE THE EXISTING CORPUS
Presets must be assembled from Roomio's existing archetype library (sofas, beds, tables, chairs, storage, decor, placeholder box) plus materials/colors. Where a genre calls for an item not yet modeled (e.g. a figurine display case for "anime"), use the closest archetype or the placeholder box, and log a request in `roomio.txt` for the new asset rather than blocking.

Personalization angle

The genres deliberately map to *who the user is / who they live with* (life-stage personas like bachelor, couple, family) and *what they love* (interest personas like anime, sports, gamer). The intake can be a simple picker: "Who's this room for?" and "What's the vibe?" Pinterest's own 2026 trend data confirms users increasingly want spaces that "tell stories and reflect personality and passion" rather than follow neutral, prescriptive rules — persona-first presets ride exactly that shift.

3. The 10 Genres (research-seeded)

Ten starting genres, split between life-stage personas and interest/aesthetic personas. The descriptions below are **research-seeded starting points** from design sources and Pinterest trend data — Agent A must verify and enrich each by looking at real pins (per §1).

#	Genre	Who it's for	Recurring elements to verify on Pinterest
1	Bachelor / Solo Pad	Single person, one occupant	Statement sofa, media/TV zone, bar nook, leather + dark tones, minimal but bold; indulgent single-user layout (no compromise pieces).
2	Young Couple	Two adults, no kids	Shared seating for two, balanced "his + hers" palette, cozy conversation zone, layered textures, one shared focal point (art/fireplace/TV).
3	Family with Kids	Parents + children	Durable/relaxed "family room / media room" feel, large sectional, soft rug, storage for toys, kid-safe layout, TV-centric, generous walkways.
4	Anime / Otaku	Anime & manga fans	Glass figure display cabinets, floating shelves of figures/Nendoroids, bold accent color, LED/neon strips, custom neon sign, character wall art/ posters, manga shelving, gaming+anime desk.
5	Gamer Setup	PC/console gamers	Desk + monitor wall, RGB/LED lighting, gaming chair, cable management, dark walls, accent neon; media zone for console.
6	Sports Fan	Football/sports lovers	Team-color accent palette, big-screen media wall, comfy lounge seating, memorabilia display shelf, bar/snack nook for game days.
7	Afrohemian	Culture-forward, boho-warm	(Pinterest 2026, +220% search) Woven raffia, mudcloth textiles, carved wood, raw ceramics, handwoven baskets; earthy reds, ochre, muted green; artisan rug as anchor.
8	Neo Deco / Glam	Bold, opulent taste	(Pinterest 2026) Geometric patterns, brass/ chrome finishes, velvet upholstery, jewel tones (emerald, sapphire, champagne), sunburst mirror, marble accents.
9	Celestial Calm	Soft, restorative seekers	(Pinterest 2026 "opal aesthetic") Iridescent/ pearlescent finishes, ivory/pearl/lavender palette, soft shimmer, star-like lighting, layered with natural materials.
10	Biophilic / Plant	Nature lovers, calm-focused	Abundant potted plants, natural wood, jute/wool textures, neutral-organic palette, large organic-shaped rug, soft daylight-oriented layout.

GENRE SET IS A STARTING POINT

These ten cover the requested life-stage personas (bachelor, couple, family) plus high-signal interest and 2026-trending aesthetics. If Pinterest research reveals a clearly stronger genre for the target audience, swap it in — but keep the count at ten and keep at least the three life-stage personas. Log any swap in `roomio.txt`.

4. How to Build a Preset (Pinterest method)

A repeatable procedure so every genre is built the same evidence-based way:

1. **Gather.** For the genre, pull 15–30 representative rooms from Pinterest boards + design sources for that exact aesthetic. Favor pins with high engagement and recent dates.
2. **Tally recurrence.** Across those rooms, count which elements appear repeatedly: furniture types, dominant colors, materials/finishes, signature decor, lighting style, layout pattern. The items that recur in most pins are the genre's "DNA."
3. **Map to corpus.** Translate each recurring element to a Roomio archetype + material + color. Missing items → closest archetype or placeholder box + a `roomio.txt` asset request.
4. **Lay it out.** Place the pieces using the §6 design rules (focal point, rug anchoring, conversation zone, walkways). The preset should itself pass the suggestion engine with no missing essentials.
5. **Encode.** Write the preset as JSON (§7) so it loads into the editor like any saved design.
6. **Caption.** Add a 1–2 sentence style note in the genre's own language.

QUALITY BAR

A preset is good if a fan of that genre would look at it and say "yes, that's the vibe" — because it's built from the elements that actually recur across that community's pins, not from assumptions.

5. Feature 2: The Design-Suggestion Engine

> As the user builds (from blank or from a preset), a lightweight engine checks the room against a **necessities rulebook** and surfaces gentle, confirmable suggestions: "This room has no rug — add one?" or "Only one light source — add a lamp?" Presented as a dismissible dropdown of suggestions, never auto-applied.

Behavior

- Runs whenever the scene changes (item added/removed/moved).
- Each fired rule produces a suggestion card: what's missing/off, why it matters (one line), and a one-tap "Add"/"Apply" that drops the suggested archetype (user can then position it).
- Suggestions are **advisory**: dismissible, never forced, never blocking. A user can ignore all of them.
- Prioritize: functional necessities (a bedroom with no bed) rank above stylistic polish (add a second light layer).

Two tiers of rules

- **Necessity rules** — functional gaps a room of this type shouldn't have (bedroom missing a bed; seating area missing seating; no lighting at all).
- **Polish rules** — documented design principles that make a room feel intentional (rug to anchor seating, layered lighting, a clear focal point, a plant for life, not pushing all furniture against walls).

Both tiers come from the rulebook in §6, which is seeded from interior-design sources and must be expanded by Agent A via further research.

6. The Necessities Rulebook (seed rules)

A starter set of rules drawn from interior-design sources. Each rule = a trigger condition + a suggestion + a rationale. Agent A should expand this via research and tune thresholds against Roomio's archetypes.

ID	Tier	Trigger (if...)	Suggest / rationale
R1	Necessity	Room typed as bedroom has no bed	Suggest a bed. A bedroom's primary function piece is missing.
R2	Necessity	Seating/living area has no seating	Suggest a sofa/chairs. No place to sit.
R3	Necessity	Room has zero light sources	Suggest a ceiling light or lamp. Every room needs light.
R4	Polish	Seating present but no rug	Suggest a rug sized to anchor the seating (front legs of sofa/chairs on it). A too-small or absent rug is the "#1 design crime"; rugs anchor and define the zone.
R5	Polish	Only one (overhead) light source	Suggest task/accent lighting (lamp, sconces). A single ceiling fixture is harsh; layer ambient + task + accent.
R6	Polish	No clear focal point in the room	Suggest anchoring around one focal point (statement sofa, art wall, media unit). Without one, a room feels directionless.
R7	Polish	All furniture pushed against walls	Suggest floating seating inward to form a conversation zone. Wall-hugging makes a room feel cold and disconnected.
R8	Polish	No greenery anywhere	Suggest a potted plant. Adds life/biophilic warmth; recurring in most well-styled rooms.
R9	Polish	Living area has a sofa but no coffee/side table	Suggest a coffee or side table. Completes the functional seating group.
R10	Polish	Walkway/clearance below threshold (furniture too dense)	Suggest spacing out for traffic flow. Leave breathing room; don't overcrowd.
R11	Polish	>1 strong color beyond a 60-30-10 balance	Suggest reining color toward 60% dominant / 30% secondary / 10% accent. Too many competing colors = visual chaos.
R12	Polish	Bed present but no nightstand/bedside lighting	Suggest a bedside table + lamp. Standard functional pairing for a bedroom.

MAKE IT GENRE-AWARE (STRETCH)

Rules can be conditioned on the active genre: an "anime" room with no figure-display shelf, or a "gamer" room with no desk/monitor, can fire genre-specific suggestions. Keep these as a separate, optional rule group layered on the universal rules above.

7. Data Schemas

Preset schema (one per genre)

```
{
  "genre_id": "anime_otaku",
  "display_name": "Anime / Otaku",
  "persona_type": "interest",          // life_stage | interest | aesthetic
  "style_note": "Figure shelves, neon glow, character art \u2014 your otaku corner.",
  "room": { "shape": "rectangular", "width_cm": 360, "length_cm": 420 },
  "materials": { "wall_color": "#1b1f2a", "floor": "dark_laminate" },
  "placed_items": [
    { "archetype_id": "bookcase", "x": .., "z": .., "rot": .., "color": ".." },
    { "archetype_id": "office_chair", "x": .., "z": .., "rot": .. }
    // ... built from RECURRING pinterest elements, mapped to corpus
  ],
  "pinterest_sources": ["board/pin refs used"],   // provenance
  "asset_requests": ["figure_display_case"]        // missing → roomio.txt
}
```

Rule schema (one per rule)

```
{
  "rule_id": "R4",
  "tier": "polish",                    // necessity | polish
  "trigger": "has(seating) && !has(rug)",
  "suggest_archetype": "rug",
  "message": "Add a rug to anchor your seating.",
  "rationale": "An absent/too-small rug leaves the zone undefined.",
  "priority": 2,
  "genre_scope": "all"                 // all | [genre_ids]
}
```

8. Functional Requirements & Acceptance

ID	Requirement	Feature
P-1	"Start from a style" entry point lists the 10 genres with preview + style note.	Presets
P-2	Selecting a genre loads a fully furnished, editable room (structure + items + materials + decor).	Presets
P-3	Every preset is built from recurring elements observed across real Pinterest pins (provenance recorded).	Presets
P-4	Presets use only existing corpus archetypes; missing items use placeholder + roomio.txt asset request.	Presets
P-5	Each preset itself passes the suggestion engine with no outstanding necessity gaps.	Both
P-6	Suggestion engine re-evaluates rules on every scene change.	Suggestions
P-7	Fired rules show as dismissible suggestion cards with one-tap Add/Apply.	Suggestions
P-8	Necessity rules rank above polish rules; suggestions never block or auto-apply.	Suggestions
P-9	Rulebook is data-driven (JSON), expandable, with rationale per rule.	Suggestions
P-10	Optional: genre-aware suggestions layered on universal rules.	Both

Acceptance test (the "done" bar)

From the start screen, pick "Anime / Otaku" → a furnished, on-vibe room loads that an anime fan recognizes as authentic (because it's built from recurring pins). Start deleting things: remove the rug → a polish suggestion appears; remove all lighting → a higher-priority necessity suggestion appears. Tapping "Add" drops the suggested archetype. All suggestions are dismissible and never block building. Repeat for a life-stage genre (e.g. "Family with Kids") and confirm the preset and rules behave.

9. Coordination & Milestones

Coordination (you are Agent A)

- This is front-end work in your own domain — but log it so Agent B stays in sync via `roomio.txt` (append-only, prefix `[YYYY-MM-DD HH:MM] [AGENT-A]`).
- When a preset needs an archetype that doesn't exist, post a `REQUEST -> ASSET` entry so it can be modeled; meanwhile use the placeholder box.
- Publish the preset + rule schemas to `/shared` if Agent B's work (detection results, camera paths) should interoperate with presets later.
- Re-read `roomio.txt` at the start of each work cycle.

Milestones

Milestone	Deliverable	Definition of done
A0	Preset + rule schemas; "Start from a style" entry point shell	Schemas committed; entry point lists 10 genres (stubbed).
A1	Pinterest research pass for all 10 genres (recurrence tally + provenance)	Each genre has a documented recurring-element list.
A2	3 presets fully built (1 life-stage, 1 interest, 1 trend) end-to-end	Load → furnished editable room that passes the engine.
A3	Remaining 7 presets built	All 10 load and look on-vibe.
A4	Suggestion engine + seed rulebook (R1–R12)	Rules fire on scene change; cards with Add/dismiss.
A5	Rulebook expansion + necessity/polish prioritization + genre-aware rules	Suggestions ranked; optional genre rules layered.
A6	Polish, asset requests resolved, provenance recorded	Presets cite pin sources; placeholders swapped where assets arrived.

> **One-line summary.** Build 10 Pinterest-grounded persona room presets (life-stage + interest + 2026-trend aesthetics), each assembled from elements that actually recur across real pins and mapped to Roomio's corpus, plus a data-driven suggestion engine that flags missing necessities and design-polish gaps as dismissible dropdown cards as the user builds. Ground everything in real pins and design sources — never intuition.

End of brief · Roomio Persona Rooms & Design Suggestions (Agent A) · Verify all genre patterns against real Pinterest pins before encoding.